SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF VERIFICATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. VCHP System Information

Procedures for verification of VCHP compliance credit eligibility are described in the Energy Code Reference Appendices Section RA3.4.4.

01	SC System ID/Name from LMCC	
02	SC System Description of Area Served	
03	Conditioned Floor Area Served by the System (ft^2)	
04	Status: Refrigerant charge verification from MCH-25	
05	Verification: Is conditioned airflow supplied to all habitable rooms in accordance with the procedure in RA3.1.4.1.7?	
Notes:		

B. VCHP Indoor Unit Information

Ducted indoor units are required to be certified to the Energy Commission as low static systems, and included in the list of certified indoor units published on the Energy Commission website at the following URL: https://www.energy.ca.gov/rules-and-regulations/building-energy-efficiency/manufacturer-certification-building-equipment.

01	02	03	04	05	06	07	08	09
Indoor Unit Name or Description of Area Served	Installed Indoor Unit Type	Indoor Unit Duct Status	Conditioned Floor Area Served By The Indoor Unit (ft^2)	Number of Air Filter Devices on Indoor Unit	Indoor Unit Required Minimum System Airflow Rate (cfm)	Status: Airflow Rate Verification from MCH-23	Is Field Verification of Default Non-Continuous Fan Operation Required?	Verification: Is Ducted Low Static Indoor Unit Certified to CEC?
Notes:								

C. Verification: Ducted Indoor Units Located Entirely in Directly Conditioned Space - RA3.1.4.3.8

Ducted indoor units shall be verified in accordance with the Verified Low Leakage Ducts in Conditioned Space procedure in Section RA3.1.4.3.8.

01	02	03	04
Indoor Unit Name or Description of Area Served	A Visual Inspection Shall Confirm the Space Conditioning Distribution System Location(RA3.1.4.1.3)	Measured Duct Leakage to Outside (cfm) Using RA3.1.4.3.4	Compliance Statement:
Notes:			

D. Verification: Ductless Indoor Units Located Entirely in Directly Conditioned Space - RA3.1.4.1.8

A visual inspection shall confirm that ductless indoor units are located entirely in conditioned space in accordance with the procedures of RA3.1.4.1.8.

01	02	03
Indoor Unit Name or Description of Area Served	Indoor Unit Installation Location Verification	Compliance Statement:
Notes:		

E. Verification: Wall Mounted Thermostats - RA3.4.5

Field verification according to the procedure in RA3.4.5 shall confirm that VCHP space conditioning zones that are greater than 150 ft^2, are controlled by a permanently installed wall-mounted thermostat.

01	02	03	04	05
Indoor Unit Name or Description of Area Served	Is a Wall-mounted Thermostat Installed in the Zone Served by the Indoor Unit?	Does the Thermostat Control the Zone's Indoor Unit?	Is the Thermostat Mounted Permanently to the Wall?	Compliance Statement:
Notes:				

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

F. Verification: Non-Continuous Fan Operation - RA3.4.6

If the certificate of compliance indicates non-continuous indoor unit fan operation was specified for compliance credit, then the system shall be field verified in accordance with the procedures in SC3.4.6 to confirm that the installed system's indoor unit + outdoor unit combination does not operate the fan continuously when the system thermostat is not calling for conditioning.

01	02	03	04	05
Indoor Unit Name or Description of Area Served	Is Non-Continuous Default Fan Operation Shown in CEC Certification Listings?	Does Indoor Unit Air Distribution Fan Operate When There Is No Call For Heating?	Does Indoor Unit Air Distribution Fan Operate When There Is No Call For Cooling?	Compliance Statement:
Notes:				

G. Verification: Installed Air Filter Sizing and Pressure Drop - RA3.1.4.7 and RA3.1.4.8

Nominal 2-inch or greater depth air filters shall be sized by the system designer to accommodate a maximum allowable clean-filter pressure drop of 0.1 inch w.c at the air filter's design airflow rate as verified according to the procedures in RA3.1.4.8. Nominal one-inch minimum depth air filters shall be allowed if the filter face area is sized based on a maximum face velocity of 150 ft per minute at the air filter design airflow rate according to the procedures in RA3.1.4.7. In order to inform the occupant of the airflow and clean filter pressure drop performance required for replacement air filters, the installer shall place a sticker in or near the filter grille displaying the air filter design airflow rate and the maximum allowed clean filter pressure drop at the design airflow rate as required by Standards Section 150.0(m)12Biv.

01	02	03	04	05	06	07	08	09	10	11	12
Indoor Unit Name or Description of Area Served	Air Filter Name or Description of Location	Air Filter Device Type	Design Airflow Rate for Air Filter Device (cfm)	Air Filter Nominal Depth (inch)	Air Filter Nominal Length (inch)	Air Filter Nominal Width (inch)	Air Filter Calculated Nominal Face Area ($inch^2$)	Air Filter Required Minimum Face Area ($inch^2$)	Face Area Compliance	Air Filter Rated Pressure Drop at Design Airflow Rate (inch W.C.)	Air Filter Pressure Drop Compliance
Notes:											

H. VCHP System Compliance Statement

01	

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified ~~HERS~~ECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require ~~HERS~~ECC verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (LMCI) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (LMCC) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building.

BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION

Company Name (Installing Subcontractor, General Contractor, or Builder/Owner):	
Responsible Builder/Installer Name:	CSLB License:

~~HERS~~ECC PROVIDER DATA REGISTRY INFORMATION

Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable):

~~HERS~~ECC RATER INFORMATION

~~HERS~~ECC Rater Company Name:	
Responsible Rater Name:	Responsible Rater Signature:
Responsible Rater Certification Number w/ this ~~HERS~~ECC Provider:	Date Signed:

LMCV-MCH-33-H User Instructions

Section A. VCHP System Information

1. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
3. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
4. This field is filled out automatically. It is referenced from the LMCI-MCH-25 which must be completed prior to this document.
5. Perform the verification specified by RSC3.1.4.1.7 and select the value that describes the result of the verification.

Section B. VCHP Indoor Unit Information

1. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
3. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
4. Accept the default value from the LMCI, otherwise enter the conditioned floor area served by the indoor unit - a value in ft^2.
5. Accept the default value from the LMCI, otherwise enter the number of air filter devices on this indoor unit.
6. This field is filled out automatically. It is referenced from the LMCI-MCH-23 which must be completed prior to this document.
7. This field is filled out automatically. It is referenced from the LMCI-MCH-23 which must be completed prior to this document.
8. This field is filled out automatically. It is referenced from the Certificate of Compliance which must be completed prior to this document.
9. Navigate to the URL for the Manufacturer certification listings and determine whether the installed system is included in the CEC listing, then select the value that describes the result of the verification.

Section C. Verification: Ducted Indoor Units Located Entirely in Directly Conditioned Space - RA3.1.4.3.8

1. This field is filled out automatically. It is referenced from a different section of this document.
2. Select the statement that best describes the location of the ducted distribution system.
3. Enter the leakage to outside airflow determined from the RA3.1.4.3.8
4. This field is filled out automatically

Section D. Verification: Ductless Indoor Units Located Entirely in Directly Conditioned Space - RA3.1.4.1.8

1. This field is filled out automatically. It is referenced from a different section of this document.
2. Select the statement that best describes the indoor unit installation location as determined according to RA3.1.4.1.8.
3. This field is filled out automatically

Section E. Verification: Wall Mounted Thermostats - RA3.4.5

1. This field is filled out automatically. It is referenced from a different section of this document.

2. Answer yes or no to the question: Is a wall-mounted thermostat installed in the zone served by the indoor unit?
3. Answer yes or no to the question: Does the thermostat control the zone's indoor unit?
4. Answer yes or no to the question: Is the thermostat mounted permanently to the wall?
5. This field is filled out automatically

Section F. Verification: Non-Continuous Fan Operation RA3.4.6

1. This field is filled out automatically. It is referenced from a different section of this document.
2. Select the best response to the question: Is non-continuous default fan operation shown in CEC certification listings?
3. Select the best response to the question: Does indoor unit air distribution fan operate when there is no call for heating?
4. Select the best response to the question: Does indoor unit air distribution fan operate when there is no call for cooling?
5. This field is filled out automatically

Section G. Verification: Installed Air Filter Sizing and Pressure Drop - RA3.1.4.7 and RA3.1.4.8

1. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
3. This field is filled out automatically. It is referenced from the LMCI-MCH-01 which must be completed prior to this document.
4. This field is filled out automatically. It is referenced from another section on this document, or from the LMCI-MCH-01 which must be completed prior to this document.
5. Enter the nominal depth of the air filter in inches.
6. Enter the nominal length of the air filter in inches.
7. Enter the nominal width of the air filter in inches.
8. This field is filled out automatically by calculating the product of air filter length and air filter width.
9. This field is filled out automatically based on the depth of the filter.
10. This field is filled out automatically
11. Input the pressure drop at the design airflow rate from the performance data information published on the air filter label.
12. This field is filled out automatically

Section H. VCHP System Compliance Statement

1. This field is filled out automatically

A. VCHP System Information

Procedures for verification of VCHP compliance credit eligibility are described in the Energy Code Reference Appendices Section RA3.4.4.3.

01	SC System ID/Name from LMCC	<<auto filled text: referenced from LMCI-MCH-01>>
02	SC System Description of Area Served	<<auto filled text: referenced from LMCI-MCH-01>>
03	Conditioned Floor Area Served by the System (ft^2)	<<autofilled text: referenced from LMCI-MCH-01a field D03, or MCH-01d field D03>>
04	Status: Refrigerant charge verification from MCH-25	<<if the system has a registered LMCI-MCH-25 that indicates compliance with the refrigerant charge verification requirement, then text result=[System Complies with Refrigerant Charge verification Requirements]; else text result=[System does not comply. A registered MCH-25 for this system is required.]
05	Verification: Is conditioned airflow supplied to all habitable rooms in accordance with the procedure in RAC3.1.4.1.7?	<<user select from following two text values: 1:[Indoor unit complies according to the criteria in RA3.1.4.1.7] 2:[Indoor unit does not comply]

B. VCHP Indoor Unit Information

Ducted indoor units are required to be certified to the Energy Commission as low static systems, and included in the list of certified indoor units published on the Energy Commission website at the following URL: https://www.energy.ca.gov/rules-and-regulations/building-energy-efficiency/manufacturer-certification-building-equipment.

<<provide one row of data for each of the indoor units listed on the MCH-01 that are associated with the SC system ID and area served data in A01 and A02. For indoor unit information refer to MCH-01a Section G, or J; otherwise refer to MCH-01d Section H, K, L>>

01	02	03	04	05	06	07	08	09
Indoor Unit Name or Description of Area Served	Installed Indoor Unit Type	Indoor Unit Duct Status	Conditioned Floor Area Served By The Indoor Unit (ft^2)	Number of Air Filter Devices on Indoor Unit	Indoor Unit Required Minimum System Airflow Rate (cfm)	Status: Airflow Rate Verification from MCH-23	Is Field Verification of Default Non-Continuous Fan Operation Required?	Verification: Is Ducted Low Static Indoor Unit Certified to CEC?
<< auto filled text: referenced from MCH-01>> note: reference applicable values as follows: **G03 on MCH-01a for split systems, **H03 on MCH-01d for split systems, **J03 on MCH-01a for packaged systems, **either K03 or L03 on MCH-01d for packaged systems	<< auto filled text: referenced from MCH-01 if a value is available either in G04 on MCH-01a or H04 on MCH-01d, else result=n/a>>	<< auto filled text: referenced from MCH-01 if a value is available either in G05 on MCH-01a or H05 on MCH-01d, else result=n/a allowed values= *Ductless *Ducted >10ft length *Ducted ≤10ft length>>	<<reference the value on the LMCI-MCH-33 B04 as default and allow the Rater to override the default value and enter a different numeric value xxxx.x *Require the sum of the values in this column to be equal to the value in A03 as condition of completion of this doc>>	<< reference the value on the LMCI-MCH-33 B05 as default and allow the Rater to override the default value and enter a different integer value xx>>	<<if B03=Ductless, then result=n/a, else auto filled integer xxxx referenced from the LMCI-MCH-23: **MCH-23a field D02, **MCH-23b field D02, **MCH-23e field D02 **MCH-23f field D02 >>	<<**if** B03=Ductless, then result=n/a, **elseif** the system has a registered LMCI-MCH-23 that indicates compliance with the minimum airflow rate requirement, **then** text result=[System complies with the airflow rate verification requirements]; **else** text result= [System does not comply. A registered MCH-23 for this system is required.]>>	<< if B03=Ductless, then result=n/a, else if the LMCC-PRF indicates credit for [Certified Auto-Fan] operation was specified, then text value=yes, else text value=no>>	<<if B03=ductless, then result=n/a, else user select one of the following two text values: *[complies - indoor unit is certified low-static in the CEC listings] *[does not comply - indoor unit is not certified]>>

Notes:

C. Verification: Ducted Indoor Units Located Entirely in Directly Conditioned Space - RA3.1.4.3.8

Ducted indoor units shall be verified in accordance with the Verified Low Leakage Ducts in Conditioned Space procedure in Section RA3.1.4.3.8.

<<If there are no indoor units in B01, for which B03= one of the following two: 1:[Ducted >10ft length], 2:[Ducted ≤10ft length],
then display the section does not apply message;
else require one row of data for each indoor unit in B01 for which B03= one of the following two: 1:[Ducted >10ft length], 2:[Ducted ≤10ft length]>>

01	02	03	04
Indoor Unit Name or Description of Area Served	A Visual Inspection Shall Confirm the Space Conditioning Distribution System Location (RA3.1.4.1.3)	Measured Duct Leakage to Outside (cfm) Using RA3.1.4.3.4	Compliance Statement:
<< auto filled text: referenced from B01>>	<<user pick from list: **Entirely in conditioned space; or **Not entirely in conditioned space>>	<<user input, numeric xxx.x>>	<<if value in C03 is ≤ 25 cfm, and B02 = [Entirely in conditioned space]; then display text: [Complies], else display text: [Does Not Comply]>>
Notes:			

D. Verification: Ductless Indoor Units Located Entirely in Directly Conditioned Space - RA3.1.4.1.8

A visual inspection shall confirm that ductless indoor units are located entirely in conditioned space in accordance with the procedures of SC3.1.4.1.8.

<<if there are no indoor units in B01 for which B03=ductless, then display the section does not apply message,
else require one row of data for each indoor unit in B01 for which B03=ductless>>

01	02	03
Indoor Unit Name or Description of Area Served	Indoor Unit Installation Location Verification	Compliance Statement:
<< auto filled text: referenced from B01>>	<<user select one of the following 4 text values: 1:[Indoor unit mounted entirely on the surface of walls, ceilings, or floors], 2:[Indoor unit protrudes through the air barrier into unconditioned spaces], 3:[Visually confirmed: Indoor unit protrudes into indirectly conditioned spaces wholly inside both the thermal boundary and the air barrier of the dwelling], 4:[Not Visually Confirmed: Installer certifies the Indoor unit protrudes into indirectly conditioned spaces wholly inside both the thermal boundary and the air barrier of the dwelling]>>	<< **if** D02=[Indoor unit mounted entirely on the surface of walls, ceilings, or floors], **then** text value=[Complies] **elseif** D02=[Indoor unit protrudes through the air barrier into unconditioned spaces], **then** text value=[Does Not Comply]; **elseif** D02=[Visually confirmed: Indoor unit protrudes into indirectly conditioned spaces wholly inside both the thermal boundary and the air barrier of the dwelling], **then** text value=[Complies], **elseif** D02=[Not Visually Confirmed: Installer certifies the Indoor unit protrudes into indirectly conditioned spaces wholly inside both the thermal boundary and the air barrier of the dwelling], **then** text value=[Complies - ~~HERS~~ECC Sampling Not Allowed]
Notes:		

E. Verification: Wall Mounted Thermostats - RA3.4.5

Field verification according to the procedure in SC3.4.5 shall confirm that VCHP space conditioning zones that are greater than 150 ft^2, are controlled by a permanently installed wall-mounted thermostat.

<<if there are no indoor units in B01 for which B04≥150, then display the section does not apply message,
else require one row of data for each indoor unit in B01 for which B04≥150>>

01	02	03	04	05
Indoor Unit Name or Description of Area Served	Is a Wall-mounted Thermostat Installed in the Zone Served by the Indoor Unit?	Does the Thermostat Control the Zone's Indoor Unit?	Is the Thermostat Mounted Permanently to the Wall?	Compliance Statement:
<< auto filled text: referenced from B01>>	<<user select one of the following two: *yes *no>>	<<user select one of the following two: *yes *no>>	<<user select one of the following two: *yes *no>>	<<if the values in E02 and E03 and E04 all=yes, then text result in this field=[complies], else text value in this field=[Does Not Comply]>>

Notes:

F. Verification: Non-Continuous Fan Operation - RA3.4.6

If the certificate of compliance indicates non-continuous indoor unit fan operation was specified for compliance credit, then the system shall be field verified in accordance with the procedures in RA3.4.6 to confirm that the installed system's indoor unit + outdoor unit combination does not operate the fan continuously when the system thermostat is not calling for conditioning.

<<if there are no indoor units listed in B01 for which B08=yes, then display the section does not apply message,
else require one row of data for each indoor unit listed in B01 for which B08=yes>>

01	02	03	04	05
Indoor Unit Name or Description of Area Served	Is Non-Continuous Default Fan Operation Shown in CEC Certification Listings?	Does Indoor Unit Air Distribution Fan Operate When There Is No Call For Heating?	Does Indoor Unit Air Distribution Fan Operate When There Is No Call For Cooling?	Compliance Statement:
<< auto filled text: referenced from B01>>	<<user select one of the following two: **[indoor unit+ outdoor unit combination is certified non-continuous in the CEC listings] **[not certified]	<<user select one of the following two text values: **[fan does not operate between calls for heating] **[fan operates continuously]>>	<<user select one of the following two text values: **[fan does not operate between calls for cooling] **[fan operates continuously]>>	<<if all of the following three conditions are true: 1: F02=[indoor unit+ outdoor unit combination is certified non-continuous in the CEC listings], 2: F03=[fan does not operate between calls for heating], 3: F04=[fan does not operate between calls for cooling], then text result=[complies], else text result=[does not comply]>>

Notes:

G. Verification: Installed Air Filter Sizing and Pressure Drop - RA3.1.4.7 and RA3.1.4.8

Nominal 2-inch or greater depth air filters shall be sized by the system designer to accommodate a maximum allowable clean-filter pressure drop of 0.1 inch W.C. at the air filter's design airflow rate as verified according to the procedures in RA3.1.4.8. Nominal one-inch minimum depth air filters shall be allowed if the filter face area is sized based on a maximum face velocity of 150 ft per minute at the air filter design airflow rate according to the procedures in RA3.1.4.7. In order to inform the occupant of the airflow and clean filter pressure drop performance required for replacement air filters, the installer shall place a sticker in or near the filter grille displaying the air filter design airflow rate and the maximum allowed clean filter pressure drop at the design airflow rate as required by Standards Section 150.0(m)12Biv.

<<if None of the indoor units listed in B01 have a value in B03 == DuctsGT10Ft, then display the section does not apply message;
else require one row of data (each) for the quantity of air filter devices in B05 for each of the indoor units in B01 that have a value in B03 == DuctsGT10Ft.

01	02	03	04	05	06	07	08	09	10	11	12
Indoor Unit Name or Description of Area Served	Air Filter Name or Description of Location	Air Filter Device Type	Design Airflow Rate for Air Filter Device (cfm)	Air Filter Nominal Depth (inch)	Air Filter Nominal Length (inch)	Air Filter Nominal Width (inch)	Air Filter Calculated Nominal Face Area (inch^2)	Air Filter Required Minimum Face Area (inch^2)	Face Area Compliance	Air Filter Rated Pressure Drop at Design Airflow Rate (inch W.C.)	Air Filter Pressure Drop Compliance
<<auto filled from: *MCH-01a K03 *MCH-01d M03>>	<<auto filled from: *MCH-01a K04 *MCH-01d M04>>	<<auto filled from: *MCH-01a K05 *MCH-01d M05>>	<<If B05=1, then autofill value referenced in B06, elseif B05>1, then reference values from: *MCH-01a K06 *MCH-01d M06 *Require the sum of the values in this column for each indoor unit in G01 to be greater than or equal to the value in B06 as condition of completion of this doc>>	<< <<user enter numeric, xxxx>>	<<user enter numeric, xxxx>>	<<user enter numeric, xxxx>>	<<calculated value= G06*G07 >>	<<**if** G05=1, **then** calculated value=(G04 ÷ 150) * 144; **else** display text result=[specified by system designer]>>	<<if value in G09= [specified by system designer], then display text result=[specified by system designer]; elseif G08≥G09, then text result in this field=[complies], else text result in this field=[Does Not Comply]>>	<<user enter numeric value 1.5≥x.xx≥0.01>>	<<if G11≤0.1, then text result in this field=[complies], else text result in this field=[Does Not Comply]

Notes:

H. VCHP System Compliance Statement	
01	<<**if** all of the following conditions are true: A04=[System Complies with Refrigerant Charge verification Requirements] A05=[Indoor unit complies according to the criteria in RA3.1.4.1.7] B07=[System complies with the airflow rate verification requirements] B09=[complies - indoor unit is certified low-static in the CEC listings C04=[Complies] E05=[Complies] F07=[Complies] G10≠[Does Not Comply] G12=[Complies] D03=[Complies] **then** text result=[VCHP System Complies] **else** text result= [VCHP System Does Not Comply]